01_LoginPage

MainActivity.java

```
package com.example.myapplication;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText user, pass;
    Button login;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        user = findViewById(R.id.user);
        pass = findViewById(R.id.pass);
        login = findViewById(R.id.login);

        login.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                if (user.getText().toString().equals("usr") &&
pass.getText().toString().equals("123")) {
                    Toast.makeText(MainActivity.this, "Login successful",
Toast.LENGTH_SHORT).show();
                } else {
                    Toast.makeText(MainActivity.this, "Invalid credentials",
Toast.LENGTH_SHORT).show();
                }
            }
        });
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="30dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="center"
        android:text="LOGIN"
        android:textSize="30sp" />

    <EditText
        android:id="@+id/user"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Username" />

    <EditText
        android:id="@+id/pass"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:inputType="textPassword" />

    <Button
        android:id="@+id/login"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="center"
        android:text="Login" />

</LinearLayout>
```

Credentials:

Username: usr
Password: 123

02_SimpleCalculator

MainActivity.java

```
package com.example.calculator;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

public class MainActivity extends AppCompatActivity {

    EditText t1;
    Button bac;
    String firstNumber = "", operator = "";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        t1 = findViewById(R.id.t1);
        bac = findViewById(R.id.bac);

        int b[] = new int[17];

        for (int i = 0; i <= 16; i++) {
            String s = "b" + i;
            int rid = getResources().getIdentifier(s, "id", getPackageName());
            b[i] = rid;
        }

        for (int id : b) {
            Button button = findViewById(id);

            if (button != null) {
                button.setOnClickListener(new View.OnClickListener() {
                    @Override
                    public void onClick(View view) {
                        handleButtonClick(view.getId());
                    }
                });
            }
        }
```

```
    bac.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            t1.setText("");
            firstNumber = "";
            operator = "";
        }
    });
}

private void handleButtonClick(int id) {

    String txt = t1.getText().toString();

    if (id == R.id.b16) {

        if (!firstNumber.isEmpty() && !operator.isEmpty() && !txt.isEmpty()) {

            String secondNumber = txt;
            double result = 0;

            double num1 = Double.parseDouble(firstNumber);
            double num2 = Double.parseDouble(secondNumber);

            switch (operator) {
                case "+":
                    result = num1 + num2;
                    break;

                case "-":
                    result = num1 - num2;
                    break;

                case "x":
                    result = num1 * num2;
                    break;

                case "/":
                    result = num1 / num2;
                    break;
            }

            t1.setText(String.valueOf(result));
            firstNumber = "";
            operator = "";
        }

    } else if (id == R.id.b10) {
```

```
            if (!txt.contains(".")) {
                t1.setText(txt + ((Button) findViewById(id)).getText());
            }

        } else if (id == R.id.b12 || id == R.id.b13 ||
                id == R.id.b14 || id == R.id.b15) {

            if (!txt.isEmpty()) {
                firstNumber = txt;
                operator = ((Button) findViewById(id)).getText().toString();
                t1.setText("");
            }

        } else if (id == R.id.b11) {

            if (txt.length() > 0) {
                t1.setText(txt.substring(0, txt.length() - 1));
            }

        } else {

            t1.setText(txt + ((Button) findViewById(id)).getText());
        }
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:background="#000000">

    <GridLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:columnCount="4"
        android:rowCount="6"
        android:padding="5dp">

        <EditText
            android:id="@+id/t1"
            android:layout_width="match_parent"
            android:layout_height="150dp"
```

```
    android:layout_row="0"
    android:layout_column="0"
    android:layout_columnSpan="4"
    android:background="#202020"
    android:gravity="center_vertical|end"
    android:inputType="numberDecimal"
    android:textColor="#FFFFFF"
    android:textSize="40sp" />

<Button
    android:id="@+id/b1"
    android:layout_width="105dp"
    android:layout_height="75dp"
    android:layout_row="1"
    android:layout_column="0"
    android:text="1"
    android:textSize="28sp" />

<Button
    android:id="@+id/b2"
    android:layout_width="105dp"
    android:layout_height="75dp"
    android:layout_row="1"
    android:layout_column="1"
    android:text="2"
    android:textSize="28sp" />

<Button
    android:id="@+id/b3"
    android:layout_width="105dp"
    android:layout_height="75dp"
    android:layout_row="1"
    android:layout_column="2"
    android:text="3"
    android:textSize="28sp" />

<Button
    android:id="@+id/b12"
    android:layout_width="105dp"
    android:layout_height="75dp"
    android:layout_row="1"
    android:layout_column="3"
    android:text="+"
    android:textSize="28sp" />

<Button
    android:id="@+id/b4"
    android:layout_width="105dp"
```

```
    android:layout_height="75dp"
    android:layout_row="2"
    android:layout_column="0"
    android:text="4"
    android:textSize="28sp" />

<Button
    android:id="@+id/b5"
    android:layout_width="105dp"
    android:layout_height="75dp"
    android:layout_row="2"
    android:layout_column="1"
    android:text="5"
    android:textSize="28sp" />

<Button
    android:id="@+id/b6"
    android:layout_width="105dp"
    android:layout_height="75dp"
    android:layout_row="2"
    android:layout_column="2"
    android:text="6"
    android:textSize="28sp" />

<Button
    android:id="@+id/b13"
    android:layout_width="105dp"
    android:layout_height="75dp"
    android:layout_row="2"
    android:layout_column="3"
    android:text="-"
    android:textSize="28sp" />

<Button
    android:id="@+id/b7"
    android:layout_width="105dp"
    android:layout_height="75dp"
    android:layout_row="3"
    android:layout_column="0"
    android:text="7"
    android:textSize="28sp" />

<Button
    android:id="@+id/b8"
    android:layout_width="105dp"
    android:layout_height="75dp"
    android:layout_row="3"
    android:layout_column="1"
```

```
    android:text="8"
    android:textSize="28sp" />

<Button
    android:id="@+id/b9"
    android:layout_width="105dp"
    android:layout_height="75dp"
    android:layout_row="3"
    android:layout_column="2"
    android:text="9"
    android:textSize="28sp" />

<Button
    android:id="@+id/b14"
    android:layout_width="105dp"
    android:layout_height="75dp"
    android:layout_row="3"
    android:layout_column="3"
    android:text="x"
    android:textSize="28sp" />

<Button
    android:id="@+id/b0"
    android:layout_width="105dp"
    android:layout_height="75dp"
    android:layout_row="4"
    android:layout_column="0"
    android:text="0"
    android:textSize="28sp" />

<Button
    android:id="@+id/b10"
    android:layout_width="105dp"
    android:layout_height="75dp"
    android:layout_row="4"
    android:layout_column="1"
    android:text="."
    android:textSize="28sp" />

<Button
    android:id="@+id/b11"
    android:layout_width="105dp"
    android:layout_height="75dp"
    android:layout_row="4"
    android:layout_column="2"
    android:text="bp"
    android:textSize="28sp" />
```

```
        <Button
            android:id="@+id/b15"
            android:layout_width="105dp"
            android:layout_height="75dp"
            android:layout_row="4"
            android:layout_column="3"
            android:text="/"
            android:textSize="28sp" />

        <Button
            android:id="@+id/bac"
            android:layout_width="215dp"
            android:layout_height="75dp"
            android:layout_row="5"
            android:layout_column="0"
            android:layout_columnSpan="2"
            android:text="AC"
            android:textSize="28sp" />

        <Button
            android:id="@+id/b16"
            android:layout_width="215dp"
            android:layout_height="75dp"
            android:layout_row="5"
            android:layout_column="2"
            android:layout_columnSpan="2"
            android:text="="
            android:textSize="28sp" />

    </GridLayout>

</LinearLayout>
```

03_Intent

MainActivity.java

```
package com.example.login;

import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText user, pass;
    Button login;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        user = findViewById(R.id.user);
        pass = findViewById(R.id.pass);
        login = findViewById(R.id.login);

        login.setOnClickListener(v -> {
            String username = user.getText().toString();
            String password = pass.getText().toString();

            if (username.equals("user") && password.equals("123")) {
                Toast.makeText(this, "Login successful", Toast.LENGTH_SHORT).show();
                Intent i = new Intent(MainActivity.this, Homepage.class);
                startActivity(i);
            } else {
                Toast.makeText(this, "Login failed", Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="30dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="LOGIN"
        android:textSize="30sp" />

    <EditText
        android:id="@+id/user"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Username" />

    <EditText
        android:id="@+id/pass"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:inputType="textPassword" />

    <Button
        android:id="@+id/login"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Login" />

</LinearLayout>
```

Homepage.java

```
package com.example.login;

import android.os.Bundle;
import android.widget.Button;

import androidx.appcompat.app.AppCompatActivity;

public class Homepage extends AppCompatActivity {

    Button logout;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_homepage);

        logout = findViewById(R.id.logout);

        logout.setOnClickListener(v -> finish());
    }
}
```

activity_homepage.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to Homepage"
        android:textSize="25sp" />

    <Button
        android:id="@+id/logout"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Logout" />

</LinearLayout>
```

AndroidManifest.xml

Add `Homepage` inside the `<application>` tag:

```
<activity
    android:name=".Homepage"
    android:exported="false" />

<activity
    android:name=".MainActivity"
    android:exported="true">

    <intent-filter>
        <action android:name="android.intent.action.MAIN" />
        <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
```

</activity>

Credentials:

Username: user
Password: 123

Main Intent:

```
Intent i = new Intent(MainActivity.this, Homepage.class);
startActivity(i);
```